

UNNAS HUSSAIN

Senior Software Engineer | Robotics | Medical Devices | Control Systems

unnashussain.com
Email: uwh547@gmail.com
Phone: (518) 331-3146

TECHNICAL SKILLS

Programming Languages:	Python, C++, C#, C, Simulink, MATLAB, Bash, Java, SQL/MySQL, Go Lang, Arduino, Racket, Rust, ACL2, LaTeX, JavaScript, CSS, HTML, CMake, Gradle
Robotics	ROS, Unix/Linux, QNX, Gazebo, CoppeliaSim, Isaac Sim, Unity, Google Colab, OpenCV, WPILib
Tools & Systems	Git, Docker, PowerShell, OnShape, Autodesk, Solidworks, KiCAD, Icarus Verilog, PSpice, BITalino, Raspberry Pi, Jetson,

EXPERIENCE

Medtronic Inc., Boston, MA

Aug 2023 – Present

Senior Software Engineer

- Developing control software, kinematics, and ML computer vision for endoluminal robot (ELR) with continuum articulation joints for navigation in complex anatomy. Delivering medical-grade teleoperation software.
- Designed and implemented torque-based homing algorithm to autonomously straighten ELR instruments.
- Developing 6DOF Kalman Filter estimator for EM-based tip configuration of ELR system intraoperatively.
- Leading key decisions in controls and embedded software architecture as Senior Software Developer.
- Implementing multi-threaded characterizing and tuning tools for real-time control loop tuning on QNX OS.
- Leading software efforts for ground-zero projects that could be applied to various organizational units with problems that can be solved with innovative uses of robotic technologies or physics-based simulation.
- Certified in DRM Software development for physics-based simulation of tool-to-tissue interactions.
- Implemented reliable torque trajectories to test surgical claspers designs with variations on cables or tendons.

Hologic Inc, Marlborough, MA

Embedded Software Engineer

Dec 2022 – July 2023

- Implemented and updated embedded control logic for motorized and pneumatic subsystems within the ThinPrep 5000 (TP5000) system for fully automatic processing of gynecology, oncology, and Pap test samples.
- Developed slide designer to create custom laser-etched slides that are read by integrated Congex cameras.
- Designed robust TP5000 “power-on” tests to guarantee the system starts in a happy state or prompts user.

Amazon Robotics, North Reading, MA

Software Development Engineer

June 2021 – Nov 2022

- Implemented software across navigation stack for over 520K Robotic Units, including task assignment with Min Cost Flow optimizers, AprilTag-based localization, and real-time driving adjustments for human interactions.
- Developed AWS Coral interceptors to compare computational load between base and optimized algorithms.
- Owned and developed internal tools to visualize live and historical metrics of robotic equipment, including real-time state updates for over 1500 daily users (Amazon Engineers and Warehouse Associates).

Medtronic Inc., Surgical Innovations, Boston, MA

Controls Engineering Co-op

Sept 2019 – Nov 2020

- Owned full development of physics-enabled simulator of Hugo Robot-Assisted Surgeries to explore, define, and validate new clinical surgeries or indications with inverse kinematic-based pathing and log playback.
- Developed 6-joint hardware-in-the-loop testing software using Simulink to inject data into live DDS topics.
- Developed PID tuning tool in Python using PyQt5 with multi-threaded smart suggestions for tuning guidance.

EDUCATION

Johns Hopkins University, Baltimore, MD

Expected Dec 2027

M.S.E in Robotics & Autonomous Systems

Current GPA: 4.0

- Engineering for Professionals - Cohort of part-time students working in industry simultaneously
- Dynamics & Kinematics, Intro to Robotics, Mathematical Modeling, Advanced Algorithms & Data Science

- Undergraduate Research Assistance in Field Robotics Lab, configuring ROS for land, sea, and air AVs
- Head Teaching Assistant for Fundamentals of Computer Science, teaching Java and introductory OOD
- Robotics Science & Systems, Navigation, Dynamics, Computer Systems, Software Development, AI, OOD, Algorithms, Healthcare Technology, Electronics, Discrete Structures, Logic & Computation

SERVICE & VOLUNTEERING

FIRST Robotics, (Melrose, MA & Boston, MA)

Software & Electrical Mentor

2021 – Present

- Mentoring over 250 high school students to design, build, and code competition-level robot in fast-paced competitive environments, fostering a welcoming learning environment for teamwork and problem solving.
- Teaching wide range and advanced topics in STEM: controls software, software design, Java, and electrical engineering, ML pipelines, GPU-accelerated SLAM, path planning, feedback loops, and computer vision.

PATENTS & RECOGNITIONS

- | | |
|--|--|
| • SYSTEMS AND METHODS FOR CLINICAL WORKSPACE SIMULATION | <i>Patent # US 2024/0256723 A1</i> |
| • HARDWARE-INDEPENDENT ADAPTIVE AUTO HOMING ALGORITHM | <i>Provisional Patent # 64/157,090</i> |
| • ARTICULATION TUBE ASSEMBLIES WITH TORQUE-BASED INDICATIONS OF ARTICULATION ANGLE | <i>Provisional Patent # 63/983,943</i> |
| • STRATEGIES TO ADJUST PROXIMAL ROLL ANGLE OF A ROBOTIC CONTINUUM DEVICE WHILE PRESERVING DISTAL ARTICULATION SECTION TIP POSITION | <i>Provisional Patent # 63/953,627</i> |
| • SYSTEMS AND METHODS FOR SHAPE SENSING OF CONTINUUM MANIPULATORS | <i>Provisional Patent # 64/123,507</i> |
| • TRANSMISSION HOUSING FOR ROBOTIC CATHETER ARTICULATION AND TOOL INSERTION | <i>Provisional Patent # 64/122,853</i> |
| • SYSTEMS AND METHODS FOR FORCE SENSING AND ESTIMATION WITH CONTINUUM MANIPULATORS | <i>Provisional Patent # 64/122,849</i> |
| • SYSTEMS AND METHODS FOR PROVIDING PREDICTICE CATHETER PATH OVERLAYS | <i>Provisional Patent # 64/122,847</i> |
| • Medtronic Global Design for Reliability & Manufacturing (DRM) Green Belt Certification | 2026 |
| • Medtronic Science & Technology Conference presenter | 2024-2026 |
| • Medtronic Technical Contributor of the Year Award nomination | 2020 |
| • Northeastern University Honors College Dean's List | 2018 – 2021 |
| • U.S. Department of Homeland Security Gordan-CenSISS Scholarhsip | 2017 – 2018 |